

Restyle Website Application Proposal: Managing a Wardrobe and Improving Fashion Sense



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Abstract

This project proposes a web application called Restyle that helps users manage their wardrobe, plan outfits for them that they can use in the future, and ultimately increase variety and confidence in users' fashion. Users can create an account, upload clothing items with photos from their devices, and the system organizes the clothing by the detected category, color, season, and formality of the clothing. The application includes an artificial intelligence scanning feature that can scan the picture of the clothing to provide a prediction based on the given data which would auto-fill fields to save time. After users have finished building their wardrobe, users can generate outfit combinations that are given a score in an indication that could show what fits well while saving the outfits they may like. The project includes an outfit preview feature to help users visualize their generated outfit and see what the differences are. Overall, Restyle aims to reduce decision fatigue, improve organization, and make outfit planning more efficient for users.

1.0 Introduction

There are many people on a daily basis who struggle to choose what they wear, especially for those with a busy schedule or large wardrobes. Users tend to forget what clothing items they own in their closets which may lead to repetitive outfits or additional extra time spent searching their wardrobe to find something for the right occasion. This type of time-inefficient decision-making can lead to decision fatigue and frustration even though the user has plenty of clothing options available. During this time, existing wardrobe solutions often require heavy manual input in which users may have to go through a whole list of items and characteristics of the clothing which does not even provide an easy way to visualize outfit combinations before committing to them in real-life. Restyle is the solution web application that was designed to make wardrobe organization and outfit planning simpler and more efficient for users to use. The system allows users to create an account, upload files to the database to store clothing items with photos of their choice while being able to see how it is organized by attributes such as category, color, season, and formality. Restyle includes an artificial intelligence scanning feature that can predict clothing information based on the image that was given by the user in which would reduce manual data entry. After users build their virtual wardrobe, the application can generate outfit combinations based on the stored items in the database and user preferences that were selected in which the system allows users to save the outfit combinations to make sure it does not lose its spot. Restyle also provides an outfit preview feature that helps users visualize their selections based on the virtual outfits that they saved in which allows users to see it virtually before making the decisions to buy the clothing in real-life. With the combination of organization, automation, and visualization in one platform, Restyle is a very powerful tool that aims to help users make faster outfit decisions and get more value from the clothing they already own or will own as well.

2.0 Problem Statement

There is a surprising challenge when it comes to choosing outfits and keeping track of clothing items especially for people with busy schedules or extremely large wardrobes. Most users do not have an easy way to view everything they own at once and they may have forgotten items they bought or rarely wear in the first place. Even if users know what clothing they own, it can still be quite difficult to match items by season, formality, and color that matches with their chosen occasion of choice. Along with their current wardrobe, buying new clothes online can be a challenge if they're unsure how their current wardrobe complements any new items. This project addresses those issues by building a system that stores wardrobe items, helps users categorize them, and generates outfit combinations based on user preferences.

2.1 Problem Definition

The problem is that users do not have a simple, reliable system that would organize their clothing and create outfit combinations quickly. Many people rely on their memory or by physically searching through clothing in order to organize a part of their wardrobe which can be inefficient and frustrating. Existing solutions often focus on only one part of the problem such as simply storing items or suggesting outfits, but do not combine many aspects of it such as organization, automation, and visualization in a smooth workflow. Restyle defines the need for a single platform that supports wardrobe storage, clothing categorization, and outfit planning all together to create a strong system that can be efficient in helping the users organize not only their wardrobe but also their mind.

2.2 Problem Rationale

This challenge is of great significance since it is one faced by many individuals around the globe on a daily basis. Failure to monitor the clothes owned by users may result in duplication, wastage of time, or even poor utilization of the clothes due to difficulty in matching. An outfit organization system is essential since it will make the decision-making process easier for users. Restyle is significant since it saves the users from wasting time making decisions while giving them control over their choices.

3.0 What is Known About the Problem

Wardrobe management and outfitting have been studied using wardrobes via apps, recommendation engines, and computer vision techniques. Many of the approaches demonstrate that users receive positive results from being able to quickly add, easily search through, and view their clothes in ways that facilitate outfits creation. Image recognition technologies have proven useful for categorizing clothes, although the predictions may sometimes be inaccurate due to the realistic photography conditions. What Restyle makes use of in this field is a combination of practical organization and AI-powered assistance through previewing capabilities.

3.1 Research & Related Work

Existing applications and studies can contribute to the design of Restyle by providing inspiration. Clothing organization applications have proven the importance of organizing clothes, yet they suffer from the necessity to manually enter data about each item of clothing. Computer vision studies reveal the potential of applying deep learning to predict the category and color of a clothing item; however, their predictions can only be taken as advice. Recommendation engines prove that clothing combinations can be generated in a practical manner using filtering and scoring techniques.

3.1.1 Wardrobe Management Applications

Apps such as Stylebook and Cladwell let users add their clothes and sometimes build an outfit, but require users to manually tag and organize them. This demonstrates that there is a need for an easy method of viewing what one owns and building outfits from previous designs. Many users do not continue using such apps because tagging new clothes is very time-consuming. Restyle solves this problem through its ability to automatically add pictures or even scan them without much difficulty.

3.1.2 Clothing Image Recognition & Categorization

Another popular use case of computer vision involves recognizing clothes, which can be done via deep learning models to classify clothing categories and estimate their colors based on pictures. Still, accuracy depends heavily on lighting conditions, background messiness, and overall picture quality. For this reason, many applications treat predictions merely as helpful recommendations and not as definitive answers. Restyle takes a similar approach by relying on automatic filling of fields using AI-scanning technology but providing the option to correct or validate these recommendations manually after that. The scanning functionality is provided by a separate Python FastAPI microservice that processes submitted clothing pictures and provides predictions of its attributes back to the main application. Regarding category estimation, the microservice can employ PyTorch computer vision models if they are available; at the same time, a fallback strategy of estimating category based on file names is employed as well. Regarding colors, the dominant color in the garment can be estimated using OpenCV techniques, such as GrabCut and k-means clustering.

3.1.3 Outfit Recommendation Systems

Most of the time, the algorithms for recommending an outfit start from choosing clothes that satisfy certain conditions, such as the season, level of formality, and colors of clothes. While some systems rely solely on sophisticated AI techniques, most of the systems resort to scoring and heuristics to create relevant results in a short period. Such solutions are quite effective because they are relatively fast and explainable. The Restyle system uses the technique of beam search and filters together with simple scoring to produce several outfit suggestions. The proposed heuristic scoring model assesses the compatibility of the outfit based on different parameters, such as complementary and neutral color coordination, balance of clothing categories, season suitability, and coherence in the levels of formality. Then, the system generates candidate outfits using beam search by selecting the best partial combinations of clothes within the pre-defined beam width.

3.1.4 3D Visualization & Virtual Try-On Approaches

The methods of visualisation differ from rather primitive approaches of outfit preview to sophisticated algorithms used for simulating the fitting of clothes on virtual figures. However, advanced try-ons would mean sophisticated 3D models and physics that may not necessarily be achievable during the current project's development cycle. Nonetheless, even with a rather basic preview option, customers will be able to make comparative analyses of the outfits and see how they look paired with each other. Restyle focuses on creating a reliable option of previewing the outfits, and the improvement of the realism of the preview would become the subsequent stage of the project. The preview feature is implemented with the help of React, whereby rendering of the elements of an outfit occurs using React Three Fiber. Namely, the model of a male mannequin represented as a mesh group with different kinds of clothes including tops, bottoms, shoes, and outerwear in the format of GLB is imported into the 3D browser scene. When an outfit is

generated, its properties such as categories and colours are dynamically applied to the meshes of the appropriate parts of the mannequin.

4.0 Proposed Method of Solving the Problem

Restyle addresses the issue by offering a solution where a web application will store clothing items, assist with tagging those items, and offer suggestions of outfits based on tags. The users may enter items either manually or by uploading images to scan and receive recommended tags such as category and color. With the items stored within the system, it is possible to suggest outfits using different filters including season and formality. Multiple options for outfits can be returned for evaluation.

4.1 Project Overview

The workflow involved in Restyle is quite basic: log in, create a wardrobe, create an outfit, save favorites, and see your results. The wardrobe page can be used to add clothes to the wardrobe in the form of images along with some information on them, while the scan option enables you to minimize your effort to add information to the wardrobe. The outfit creation algorithm works with the information stored in the wardrobe to come up with combinations based on filter options selected by you.

4.2 Benefits, Significance, and Future

The Restyle application will provide users with convenience when selecting their outfits as it will ensure that they have an organized wardrobe, save time on outfit selection, and maximize utilization of the clothes that they already have. The application is important because it incorporates different computer science components including web development, database systems, image processing, and visualization. In addition to that, this program provides an

example of the effective implementation of AI as an assisting technology instead of replacing decisions made by a user.

5.0 Objectives

The objective of this project is to build a complete, working wardrobe and outfit-planning application that addresses real user needs. Restyle allows users to securely manage their wardrobe, generate outfit combinations using configurable filters, save preferred outfits, and visualize them on a 3D mannequin model in the browser. The project also reduces manual data entry by offering an AI-powered scanning feature that predicts clothing attributes directly from an uploaded image. Finally, the system is designed to be stable, secure, and straightforward to install and demonstrate locally.

5.1 Functional Requirements

Authentication. Users can create an account by providing a username, email address, and password of at least six characters. Returning users can sign in with their email and password. The system issues a JSON Web Token upon successful login, which is required to access all protected pages and API endpoints. Users can also log out, which clears the stored token from the browser.

Wardrobe Management. Authenticated users can add clothing items to their personal wardrobe by providing a name, category, color, formality level, season, optional notes, and an optional clothing image. Items are displayed in a card-based grid and can be deleted individually. All wardrobe data is isolated per user so that no account can access another user's clothing items.

AI Scanning. When adding a clothing item, users may upload an image and click "Scan Photo &

Fill Fields" to receive AI-generated predictions for category, color, and season. The scanning service uses OpenAI CLIP (ViT-Base) as the primary model, with a ResNet18 ImageNet fallback and filename based heuristics as a final fallback. All predictions are returned as editable suggestions so users can confirm or correct them before saving.

Outfit Generation. Users can generate outfit combinations from their wardrobe using three configurable filters: formality level, season, and outfit variety. The variety setting controls beam width, Focused produces three outfits, Balanced produces five, and More Variety produces eight. The system uses a four pass beam search algorithm that incrementally adds a top, bottom, shoes, and optional outerwear, scoring each combination and pruning to the highest-scoring candidates at each pass.

Saved Outfits. Users can save any generated outfit directly from the Outfit Generator page. Saved outfits are stored in an application-level context and viewable on the Saved Outfits page. Users can remove any saved outfit at any time.

Outfit Preview. Users can open the Outfit Preview page to visualize saved outfits on an interactive 3D mannequin rendered with React Three Fiber. Clothing meshes are color-mapped dynamically based on the stored color attribute of each item, and only the clothing pieces present in the selected outfit are rendered. Users can navigate between saved outfits and rotate the model using OrbitControls.

5.2 Non-Functional Requirements

The interface is designed around one primary task per page to keep navigation clear and avoid overwhelming users. Forms include descriptive labels and placeholder text for all clothing

attributes. Scanning results appear as pre-filled but fully editable fields rather than locked values. The application provides real-time visual feedback, including loading indicators, success confirmations, and color coded error messages, so users always know the outcome of their actions. Buttons are disabled during active operations to prevent duplicate submissions.

Outfit generation runs entirely in the Node.js backend using a lightweight beam search algorithm and completes in under 100 milliseconds regardless of wardrobe size. The 3D preview renders at 60 frames per second using Three.js's WebGL render loop. Uploaded images are resized and re-encoded by the Sharp library, before storage, keeping load times fast.

The scanning service is optional and runs as a separate FastAPI process. If the service is unavailable, the backend returns a clear error message and the main application continues to function normally, with users able to enter clothing attributes manually. All API routes return descriptive HTTP status codes and sanitized error messages without exposing internal stack traces.

5.3 Scope & MVP Definition

The implemented MVP includes:

- User authentication with JWT and bcryptjs password hashing
- Wardrobe management: add, view, and delete clothing items with image upload
- AI scanning via an optional FastAPI service using CLIP and ResNet18 for category, color, and season prediction
- Outfit generation using a four-pass beam search with color, formality, and season scoring
- Saved outfits managed through a React Context provider
- A 3D outfit preview rendered with React Three Fiber, using a GLB mannequin model with dynamically color-mapped clothing meshes

Features explicitly outside the scope of this version include full cloth physics simulation,

user-to-user sharing or social features, wardrobe pagination for very large inventories, outfit undo/restore after deletion, and photorealistic virtual try-on using the user's own image. This scope allows the team to deliver a fully functional, demonstrable system within the course timeline while laying a solid foundation for future enhancements.

6.0 Activities

Development was organized into sequential phases where each completed feature enabled the next. The first phase established the backend foundation: the PostgreSQL database schema, Express API structure, JWT authentication, and bcryptjs password hashing. The second phase built the wardrobe module, including the clothing item CRUD endpoints, multer-based image upload, and Sharp image processing. The third phase implemented the four-pass beam search outfit generation algorithm and its API endpoint. The fourth phase integrated the optional FastAPI scanning service using CLIP and OpenCV color detection, along with the frontend scanning flow on the wardrobe page. The fifth phase added saved outfits using a React Context provider. The final phase built the React Three Fiber 3D preview, loaded the GLB mannequin model, and implemented dynamic color and mesh visibility mapping.

6.1 Functionality Plan

Functionality was built in a dependency ordered sequence so each layer supported the next. Authentication and database connectivity were implemented first so that all user data could be stored securely from the start. Wardrobe management followed, covering add, view, and delete operations with image upload and Sharp processing. Outfit generation was then implemented using stored wardrobe attributes and the beam search algorithm, with formality, season, and variety filters exposed through the API. The scanning service was integrated next as an optional layer that communicates with the FastAPI process. The main application continues to

function when the service is offline. Saved outfits were added using a React Context provider to allow outfit state to persist across pages without requiring additional database writes. Finally, the React Three Fiber outfit preview was built using the GLB mannequin model, with mesh traversal to identify clothing zones and dynamic material updates for color mapping. This ordering ensured the project reached a working end-to-end state early and left time for testing and visual polish.

6.2 User Experience & UI Plan

The user interface is organized so that each page focuses on a single primary action, keeping the experience intuitive for first-time users. The Home page greets logged-in users with a time-of-day message and a weather-based outfit suggestion powered by the OpenWeather API, and scrolling reveals feature cards and a four-step guide to using the application. The Wardrobe page presents a toggleable form for adding items and a card grid for viewing existing ones, with the scanning flow integrated directly into the add form. The Outfit Generator separates filters on the left from generated results on the right, making the workflow easy to follow. The Saved Outfits page provides a focused list of saved combinations with individual remove buttons. The Outfit Preview page pairs a left-side outfit selector and item list with a right-side 3D canvas that responds immediately to outfit selection changes.

Throughout the application, user actions receive clear feedback: buttons display loading states during operations, success messages confirm completed actions, and error banners explain problems without exposing technical details. Scanning predictions populate fields automatically but remain fully editable so users always have final control. Forms validate input on the frontend before submission to catch missing required fields early.

6.3 Security & Privacy Plan

Restyle follows standard security practices appropriate for a student-level web application. Passwords are hashed using bcryptjs with 10 salt rounds before being stored in the database, and plain-text passwords are never logged or returned in any API response. The system uses JSON Web Tokens with a seven-day expiry to authenticate sessions, and the shared authenticateToken middleware enforces token verification on every protected route. All clothing data queries and outfit operations are scoped to the authenticated user's ID using parameterized SQL queries, ensuring that users can only read, modify, or delete their own data.

7.0 Development Environment

Restyle will be created in a contemporary web development environment capable of supporting rapid UI design and reliable backend functions. For the UI part, React technology will be utilized to enable reuse of UI elements and seamless page navigation. In terms of the backend part, the system will incorporate Node.js along with Express to create API endpoints responsible for user authentication, wardrobe management, and outfits creation. The relational database will be realized with PostgreSQL and will include user data, clothing items, and associations between the wardrobe, the saved items, and preview.

7.1 Software Requirements

Software Requirements include development tools for frontend, backend, database management, and an optional machine learning module. React along with its associated libraries will be used for the development of the frontend components. The backend API will use Node.js and Express. Persistence will be managed using a PostgreSQL database. The users will be able to register their accounts and add clothes to their wardrobe by either typing or through a scan service where users can upload pictures, and Python with FastAPI and image-processing libraries will be used for scans. PyTorch can be utilized for the inference of the model. The generation of

outfits will be done by using a beam search algorithm where the user sets certain filters and gets results for the selected items shown on a 3D mannequin created with Three.js and React Three Fiber.

7.2 Hardware Requirements

The requirements required by the project of Restyle fall within the limits for students carrying out the projects. The application is not necessary to have a high-end computer since it has very little hardware requirements. The software is capable of running on any of the following operating systems: Windows, Mac OS, or Linux. The laptop or computer will only be tasked to run React front end, Node back end, and PostgreSQL locally. When it comes to scanning, it will run effectively, although there are ways to improve its processing power with CPU or even GPU.

8.0 Architecture Design

The Restyle software uses the concept of modularity, where the UI, backend, database, and scanning service (in case needed) are all independent components. This makes coding easier because the code can be written separately for each component, which then only performs its intended task. The front end manages user interaction and the display of information that comes from the back end. It is the back end's responsibility to log in, authorize, and carry out other operations like making an outfit from the business logic.

8.1 System Architecture Overview

The system uses a client-server approach whereby the React front end makes calls to the backend server via REST APIs. The backend, on its part, interfaces with the PostgreSQL database to save user accounts, pieces of clothing, and outfits stored. In case the scan option is selected, the image gets sent to the FastAPI service by the backend for prediction, which in turn

is provided back to the frontend for verification by the user.

8.2 ML Scanning Service

ML Scanning service analyzes the uploaded images of clothes and outputs prediction results related to their categories and colors. To do this, FastAPI will be used to accept input images from the client via API calls and return JSON results instantly. Preprocessed images will then be analyzed using a model or method based on PyTorch technology. As predictions might not be accurate, the system considers the outputted data only suggestions for the users to review and edit their fields if needed.

8.3 Outfit Preview

Outfit preview would allow the user to view the outfit chosen by him/her rather than just read descriptions of the clothing items along with their characteristics. The outfit preview would be presented through a 3D mannequin representation or any other structure which displays all the outfit clothing items in such an organized manner that all of the clothes in the outfit will be displayed in proper color. The user can make use of the zooming facility to view the outfit properly. It has been aimed to deliver visualization to the user so that he/she can differentiate between different choices of outfits. Despite being unable to provide for realistic try-ons, the outfit preview tool would prove beneficial to the user in many ways.

8.4 Data Model Overview

The data model for Restyle is planned to be constructed in a way that allows users, clothing items, and outfits to be efficiently connected together. There would be a users table that stores users' accounts and password hashes. The clothing_items table would include all the necessary information about clothing items and their connection to the right user. An outfits table

would hold outfits created by users and additional information such as the filters used when generating outfits. Another table called outfit_items could connect outfits to their corresponding clothing items, allowing users to specify the slots (top, bottom, and shoes).

9.0 UML Diagrams

A UML diagram will be used for describing the interaction between the user and Restyle, as well as how requests are handled by the system. UML diagrams can help in conveying design choices through diagrams and help with planning implementation. The use case diagram illustrates the primary tasks that the user can perform, whereas the sequence diagram illustrates the communication process between the front-end, back-end, database, and scanning service.

9.1 Use Case Diagram

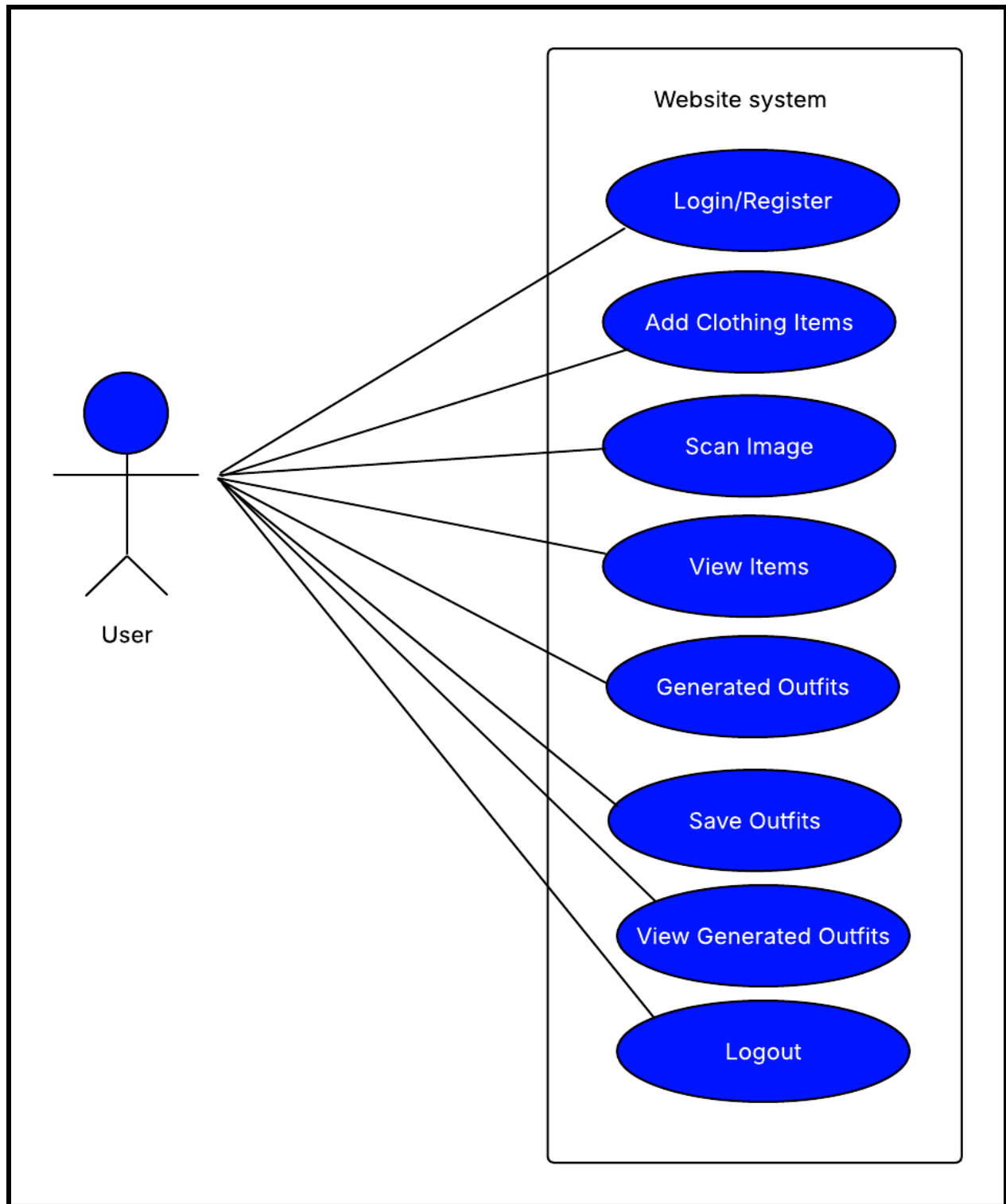


Figure 1 Use Case Diagram

9.2 Sequence Diagram

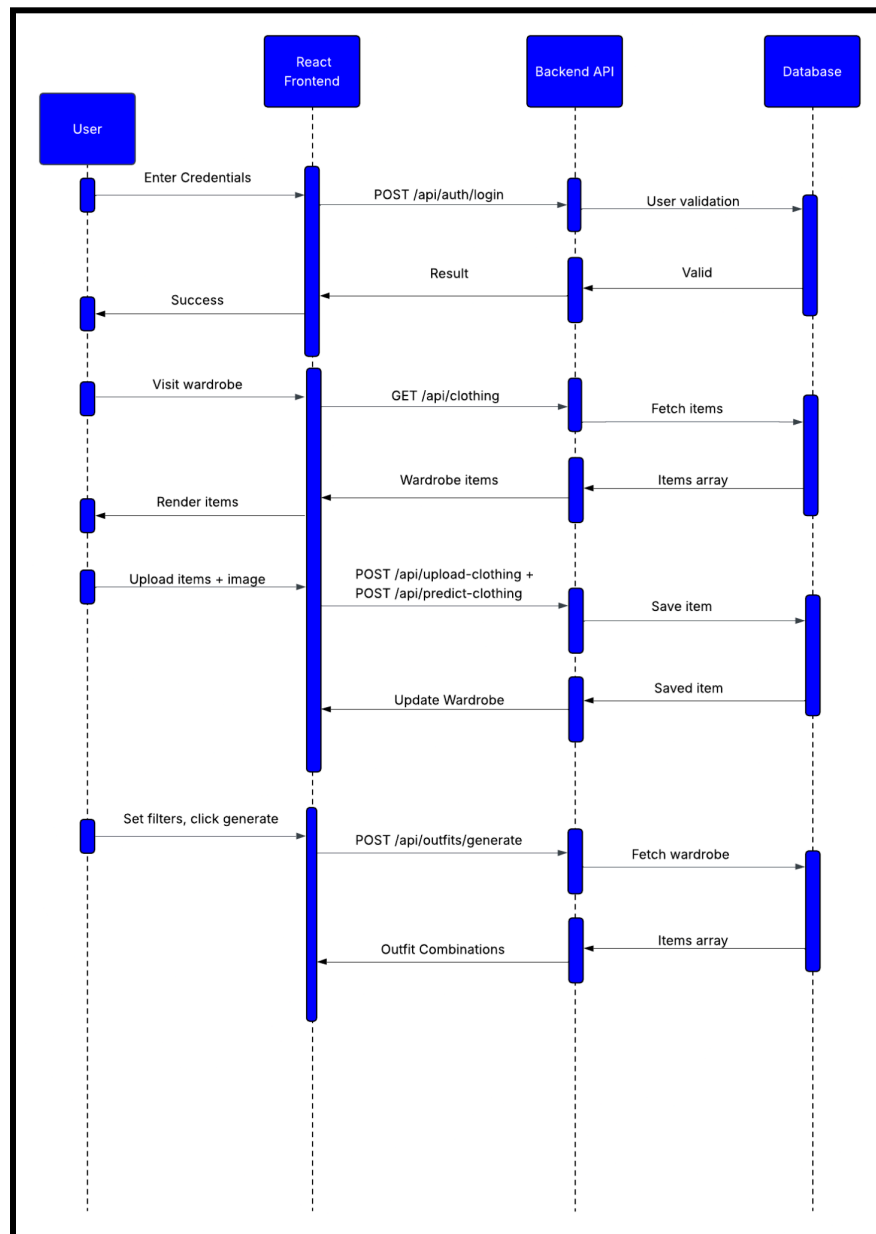


Figure 2 Sequence Diagram

10.0 Implementation

The implementation process will be done on an application page basis so that everything is documented and development proceeds in line with the user flow. The plan is to first develop authentication, then add wardrobe management and image uploads. Once we get to a good point with wardrobe management, we will add outfit generation and storing the outfits. The scanning service and viewing the outfits will be done afterward to complete everything.

10.1 Sign-in Screen

Once the user clicks on the login/signup option, he/she is redirected to the login screen. In case he/she already has an account, he/she can enter his/her email address and password. Else, he/she can click on the signup button.

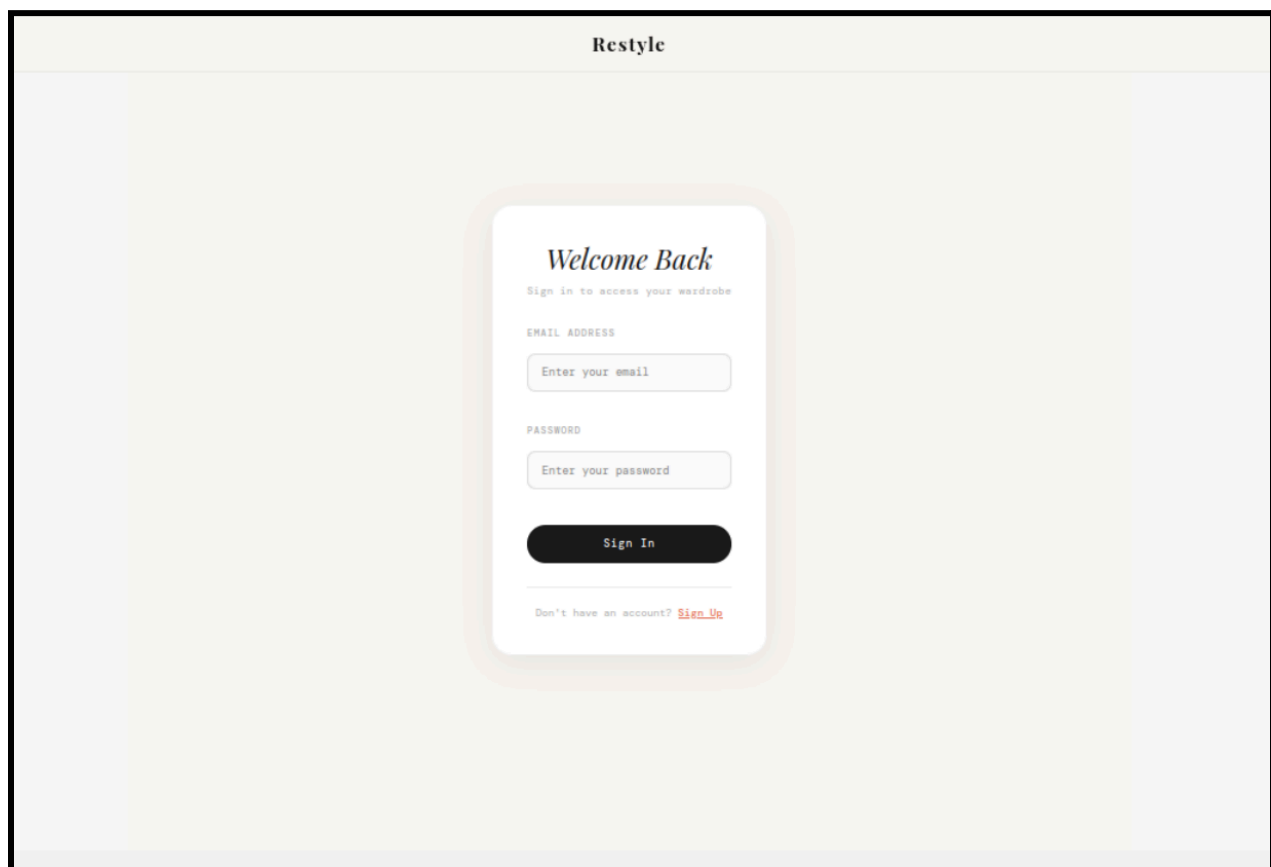
The image shows a mobile application sign-in screen for 'Restyle'. The screen has a light beige background with a darker beige header bar at the top containing the word 'Restyle' in a bold, sans-serif font. In the center, there is a white rounded rectangle with a subtle orange shadow. Inside this rectangle, the text 'Welcome Back' is written in a large, italicized, black serif font. Below it, in a smaller, lighter grey sans-serif font, is the text 'Sign in to access your wardrobe'. Further down, the label 'EMAIL ADDRESS' is followed by a text input field with the placeholder text 'Enter your email'. Below that, the label 'PASSWORD' is followed by a text input field with the placeholder text 'Enter your password'. At the bottom of the form is a dark grey rounded rectangular button with the text 'Sign In' in white. Below the button, there is a horizontal line and then the text 'Don't have an account?' followed by a red, underlined link that says 'Sign Up'.

Figure 1 Sign-in

10.2 Sign-up Screen

In case the user selects the "Sign up" option, they will be redirected to the registration page where they must fill out his username, e-mail and password. Any username and password containing 6 digits can be selected by the user. After account creation, the user will automatically be taken to the homepage.

Restyle

Create Account

Start optimizing your wardrobe today

USERNAME

Enter your name

EMAIL ADDRESS

Enter your email

PASSWORD

Enter your password

CONFIRM PASSWORD

Confirm your password

Create Account

Already have an account? [Sign In](#)

Figure 2 Sign-up

10.3 Homepage

After logging in, the user will be redirected to the homepage. The greeting for the user depends on the present time, which may either be good morning, afternoon, and evening. The user can hover the mouse pointer on the items listed, and it will display the description of that item. On scrolling down from the homepage, it will show the description of each page, and through the arrows provided, they will be directed to the chosen page. In addition, there are some guidelines that the user can follow.

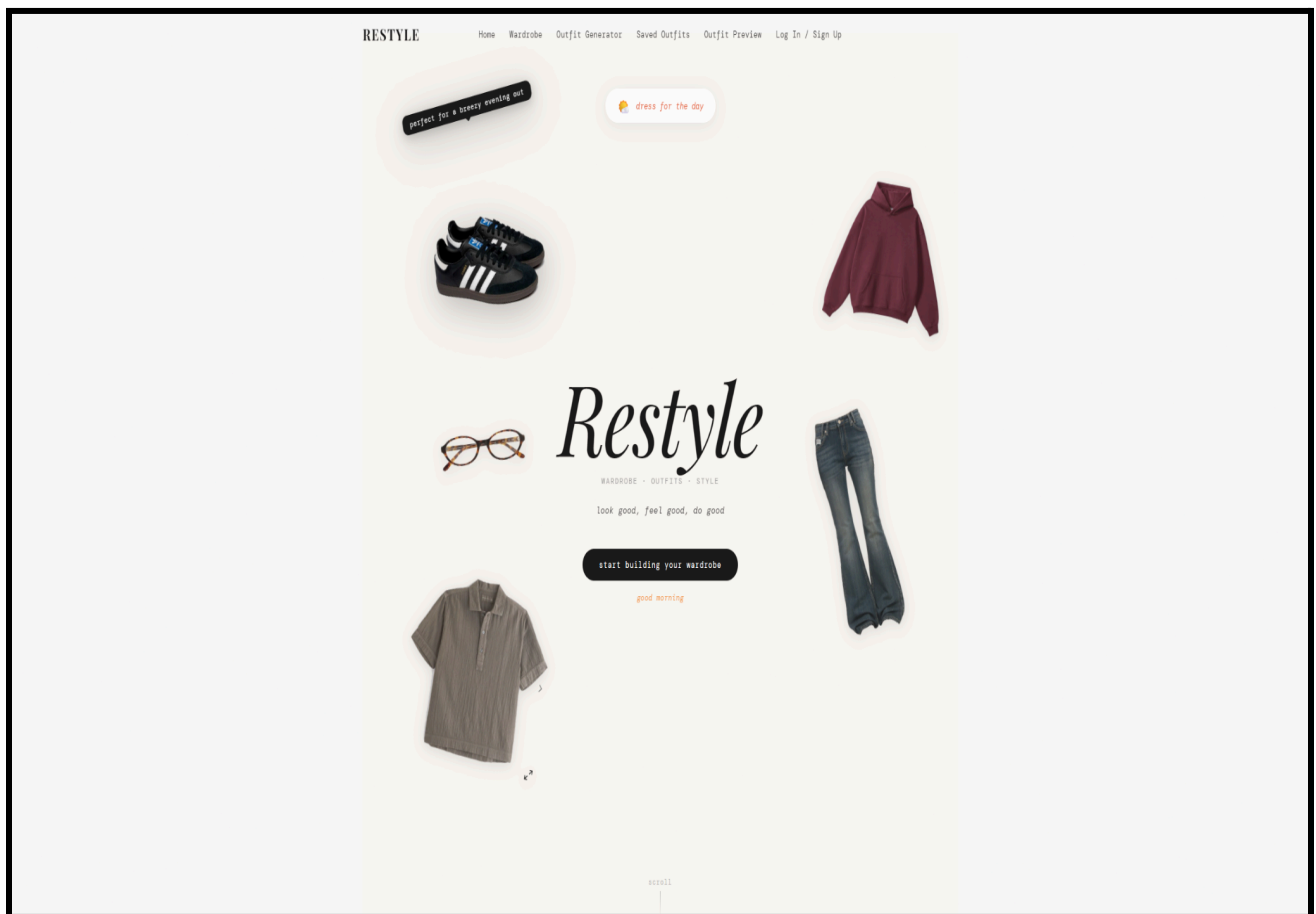


Figure 3 Homepage

10.3.1 Homepage

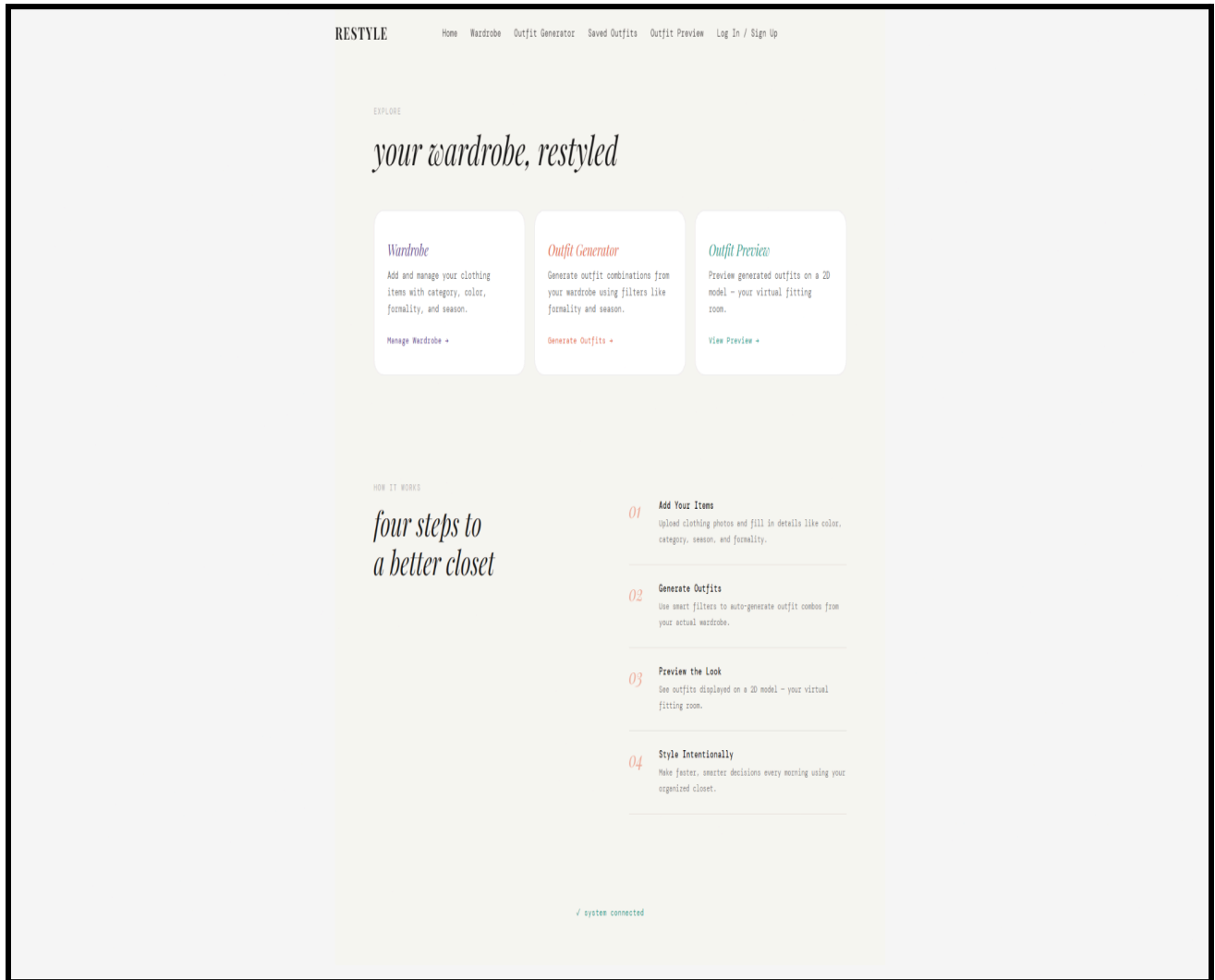


Figure 4 Homepage (Strolled-Down)

10.4 Wardrobe

Wardrobe page is the one in which the user enters the clothes and adds the item into the wardrobe. In order for the users to get their outfit combinations on the Outfit Generator page, they need to add their clothes in the wardrobe. In order to add the clothing items, they have to fill in the name, category, and color of the clothes. The users also can upload an image of the clothes they want to add and scan it using PyTorch technology. By clicking on the button Scan Photo & Fill Fields, all of the fields are filled up automatically. It does not always identify the clothes correctly so that users can manually correct every field. They can add the items after filling up every field, and then they will be shown in Your Items section.

The screenshot displays the 'My Wardrobe' interface. At the top, a navigation bar includes 'RESTYLE' and links for 'Home', 'Wardrobe', 'Outfit Generator', 'Saved Outfits', 'Outfit Preview', and a 'Logout' button. The main heading is 'My Wardrobe'. Below it is a 'Close Add Item Form' button. The 'Add New Clothing Item' form contains the following fields:

- ITEM NAME ***: A text input field with a placeholder example 'e.g., Blue Jeans, White T-shirt'.
- ITEM IMAGE**: A file upload area with a 'Choose File' button and the text 'no file chosen'.
- Scan photo & fill fields**: A button to trigger automatic field filling.
- Lightweight local scan (Python server). Optional: add PyTorch later for better type detection.**: A descriptive note below the scan button.
- CATEGORY ***: A dropdown menu currently showing 'Top (generic)'.
- COLOR ***: A text input field with a placeholder example 'e.g., Blue, Red, Black'.
- FORMALITY LEVEL**: A dropdown menu currently showing 'Casual'.
- SEASON**: A dropdown menu currently showing 'All Seasons'.
- NOTES**: A text area with a placeholder 'Any additional details about this item'.
- Add Item**: A button at the bottom of the form to submit the new item.

Below the form, a section titled 'Your Items (7)' is partially visible.

Figure 5 Wardrobe (Add)

10.4.1 Wardrobe

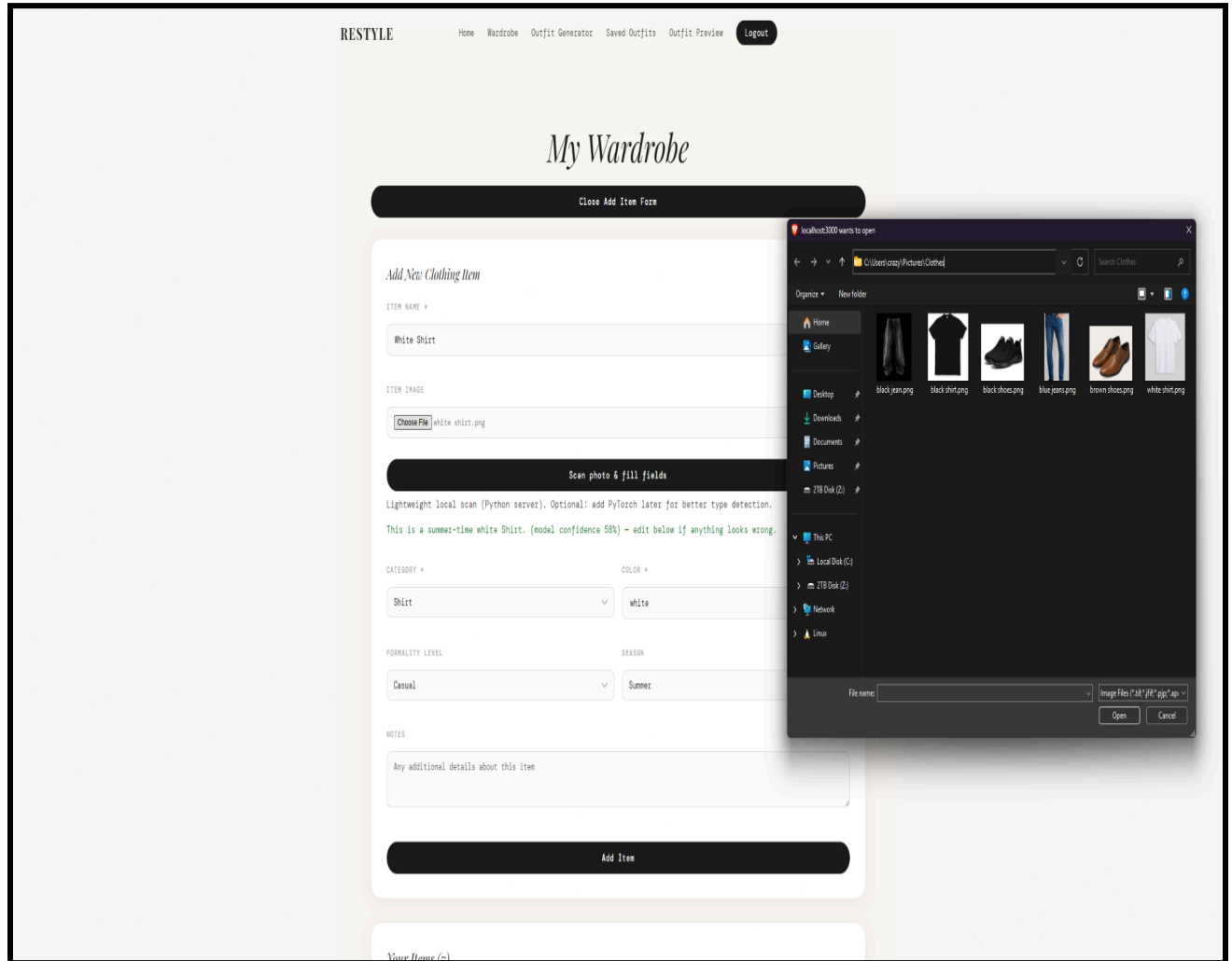


Figure 6 Wardrobe (Scan)

10.4.2 Wardrobe

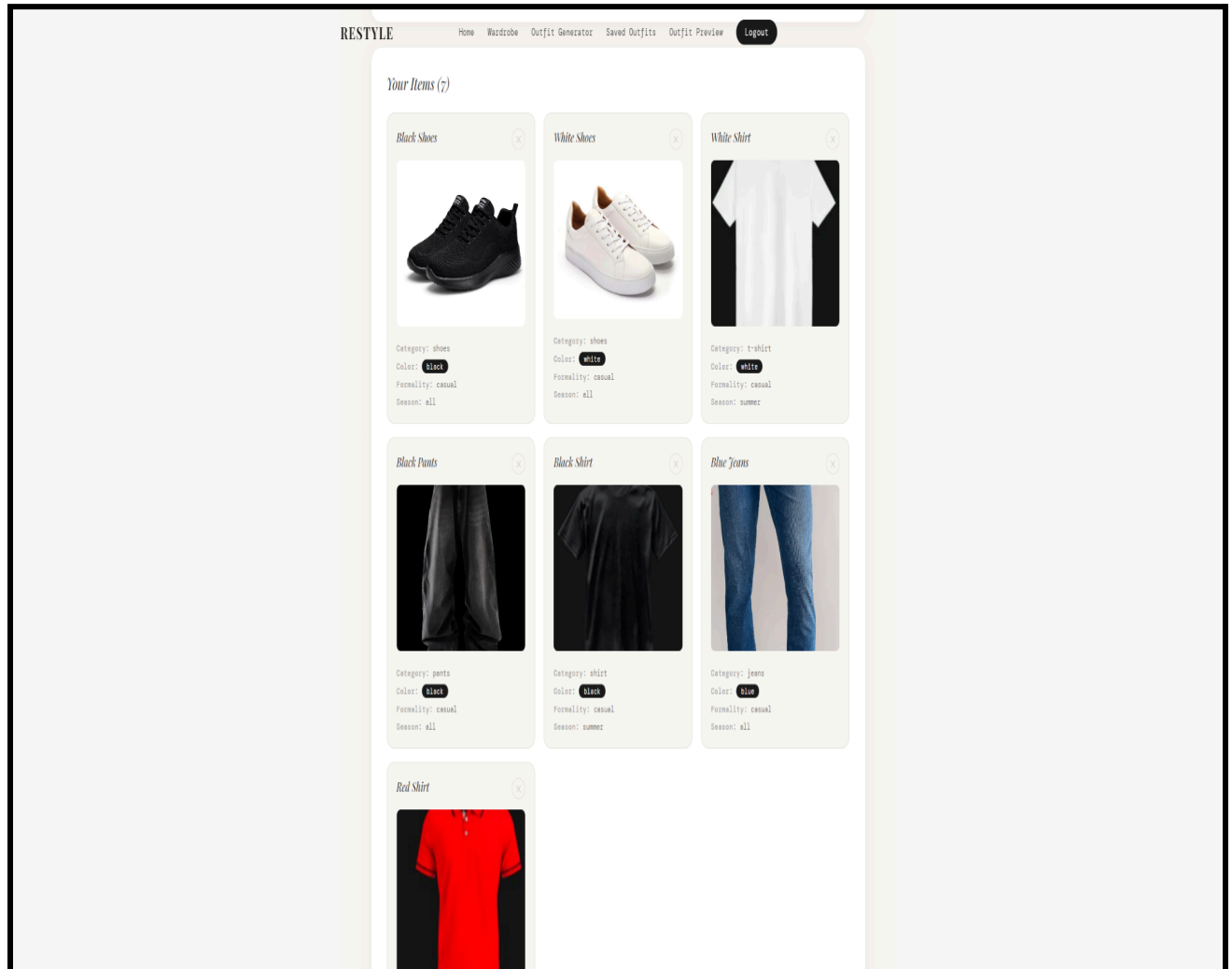


Figure 7 Wardrobe (Edit)

10.5 Outfit Generator

Outfit Generator page, like its name implies, will generate outfits through using beam search, selection heuristics and scoring methods. Prior to the generation process, the users may take advantage of outfit generation filters which will help them to customize the process of outfit generation. The first two filters are formality level and season. This depends on the information inputted about the items the user added in the wardrobe page. If the formality level and seasons of the clothes item are exactly the same with those of the generation filters, it is very possible that it will come out during the outfit generation. Lastly, there is also the outfit variety filter with three selections: focused, balanced and more variety which generates three, five and eight outfits respectively.

The screenshot shows the 'Outfit Generator' page within the 'RESTYLE' application. The navigation bar at the top includes 'Home', 'Wardrobe', 'Outfit Generator', 'Saved Outfits', 'Outfit Preview', and a 'Logout' button. The main heading is 'Outfit Generator' with the subtitle 'Generate outfit combinations from your wardrobe based on your style filters'. On the left, there are three dropdown filters: 'FORMALITY LEVEL' set to 'All Levels', 'SEASON' set to 'All Seasons', and 'OUTFIT VARIETY' set to 'Balanced'. A 'Generate Outfits' button is at the bottom of the filter panel. A large white box on the right contains the instruction: 'Select your preferences and click "Generate Outfits" to get started. Your wardrobe items will be used to build outfit combinations.'

Figure 8 Outfit Generator (Empty)

10.5.1 Outfit Generator

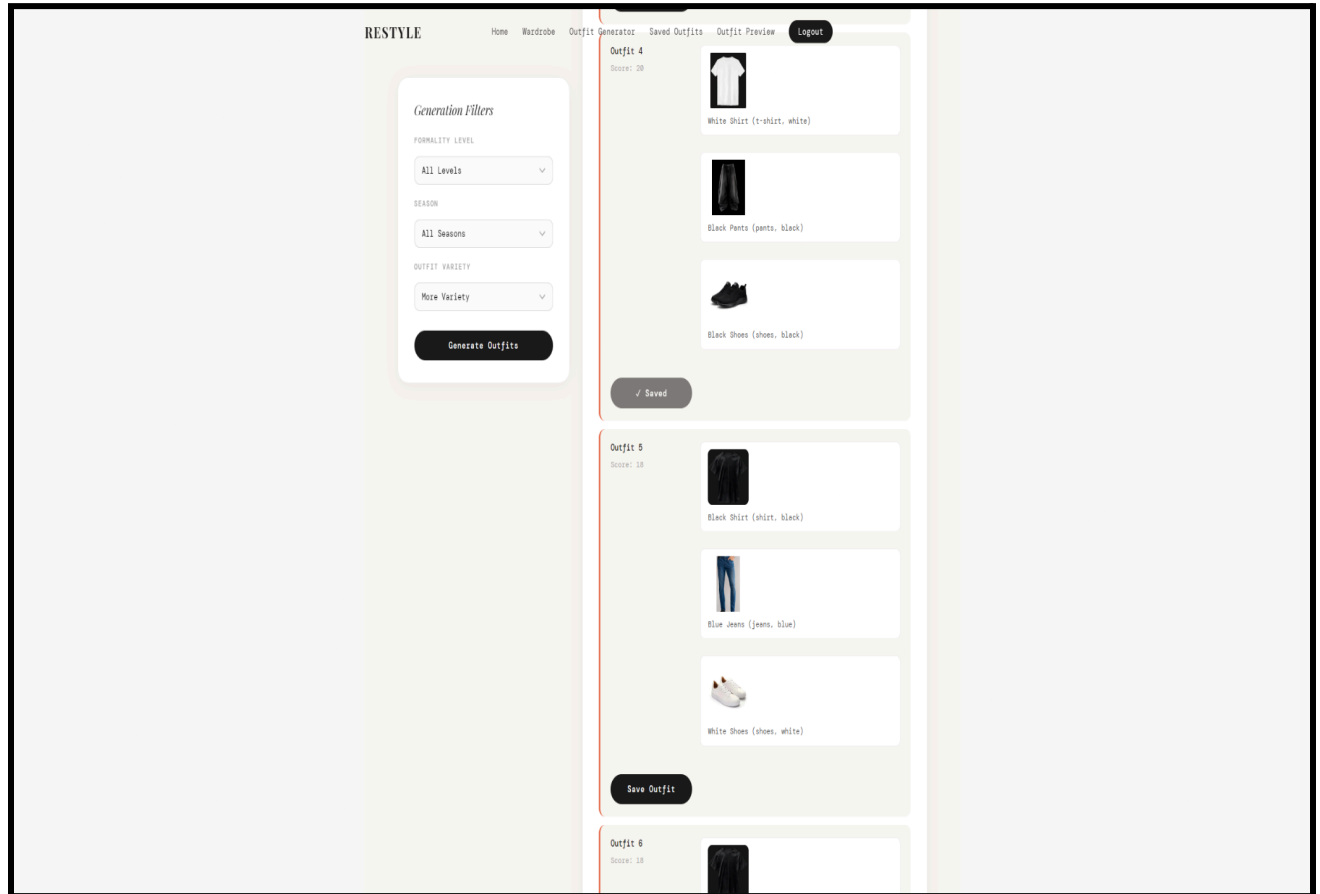


Figure 9 Outfit Generator (Full)

10.6 Saved Outfits

The Saved Outfits page is where the users can view the saved outfits they stored into the system. The user can remove any of the saved outfits.

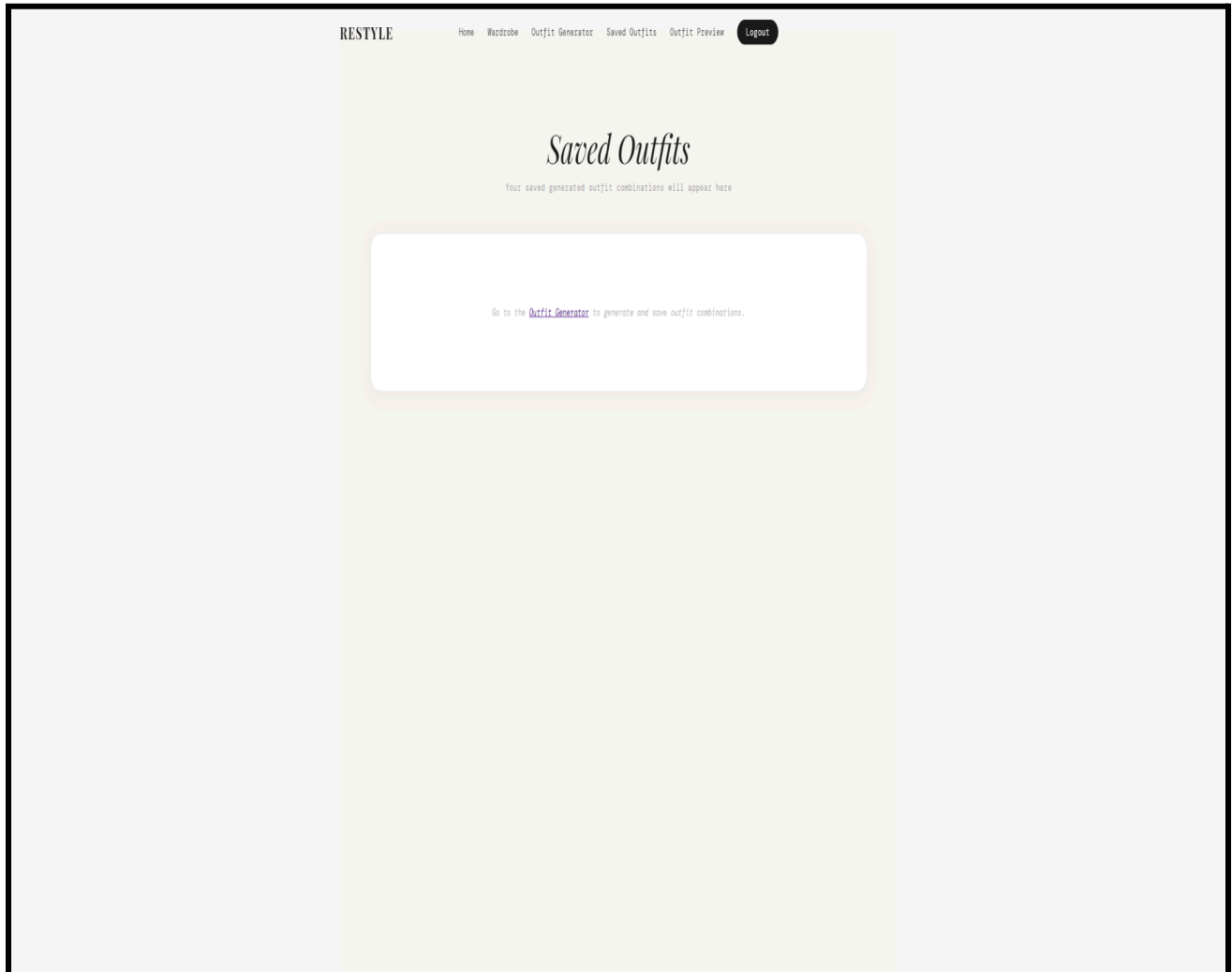


Figure 10 Saved Outfits (Empty)

10.6.1 Saved Outfits

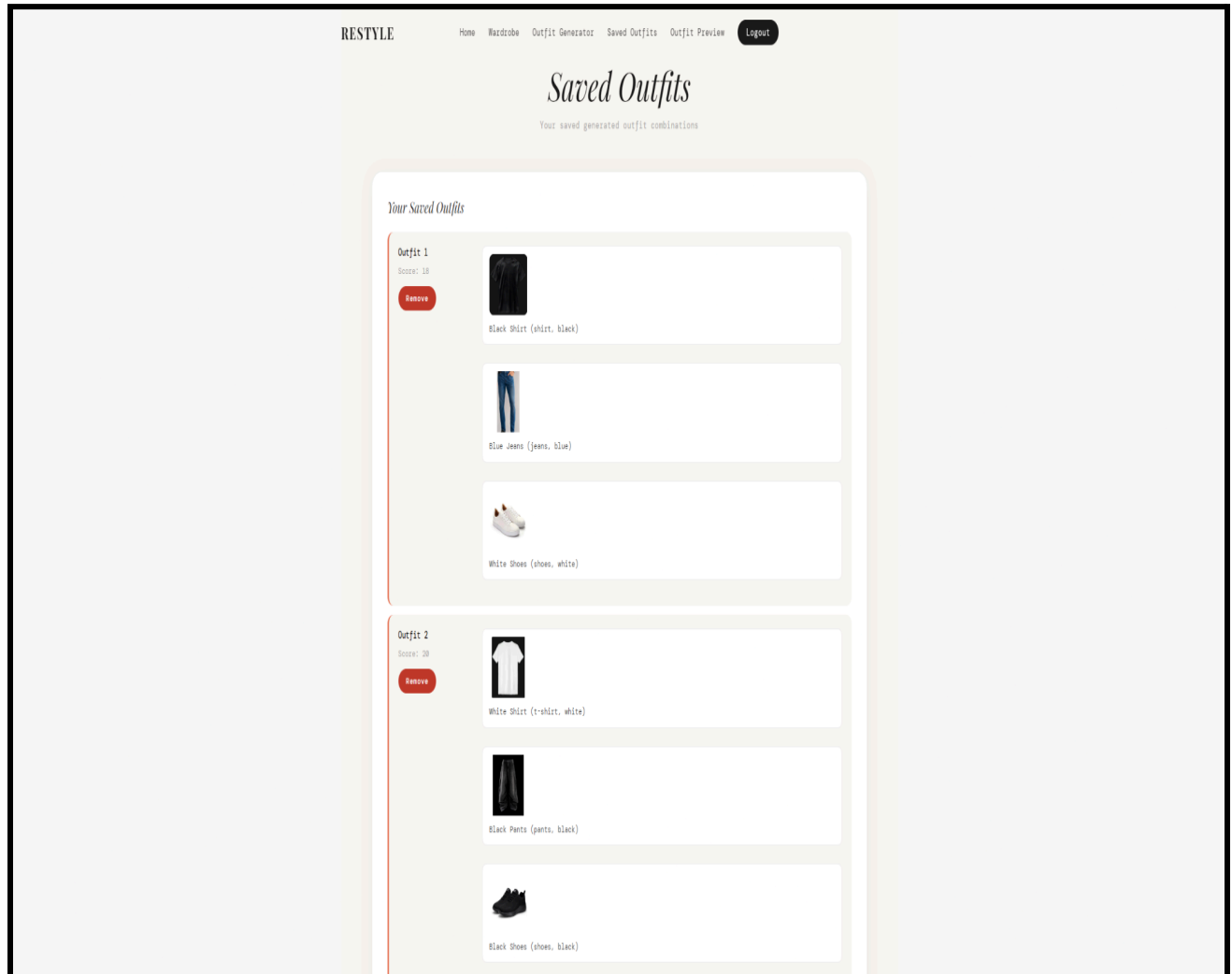


Figure 11 Saved Outfits (Full)

10.7 Outfit Preview

Preview of outfits includes the blender model to demonstrate the way the outfits generated by the generator page may appear like. This depends on the clothing color in the wardrobe page since this is where you have uploaded your saved clothing. For instance, in case you save clothing of white color, it will show white on the model.

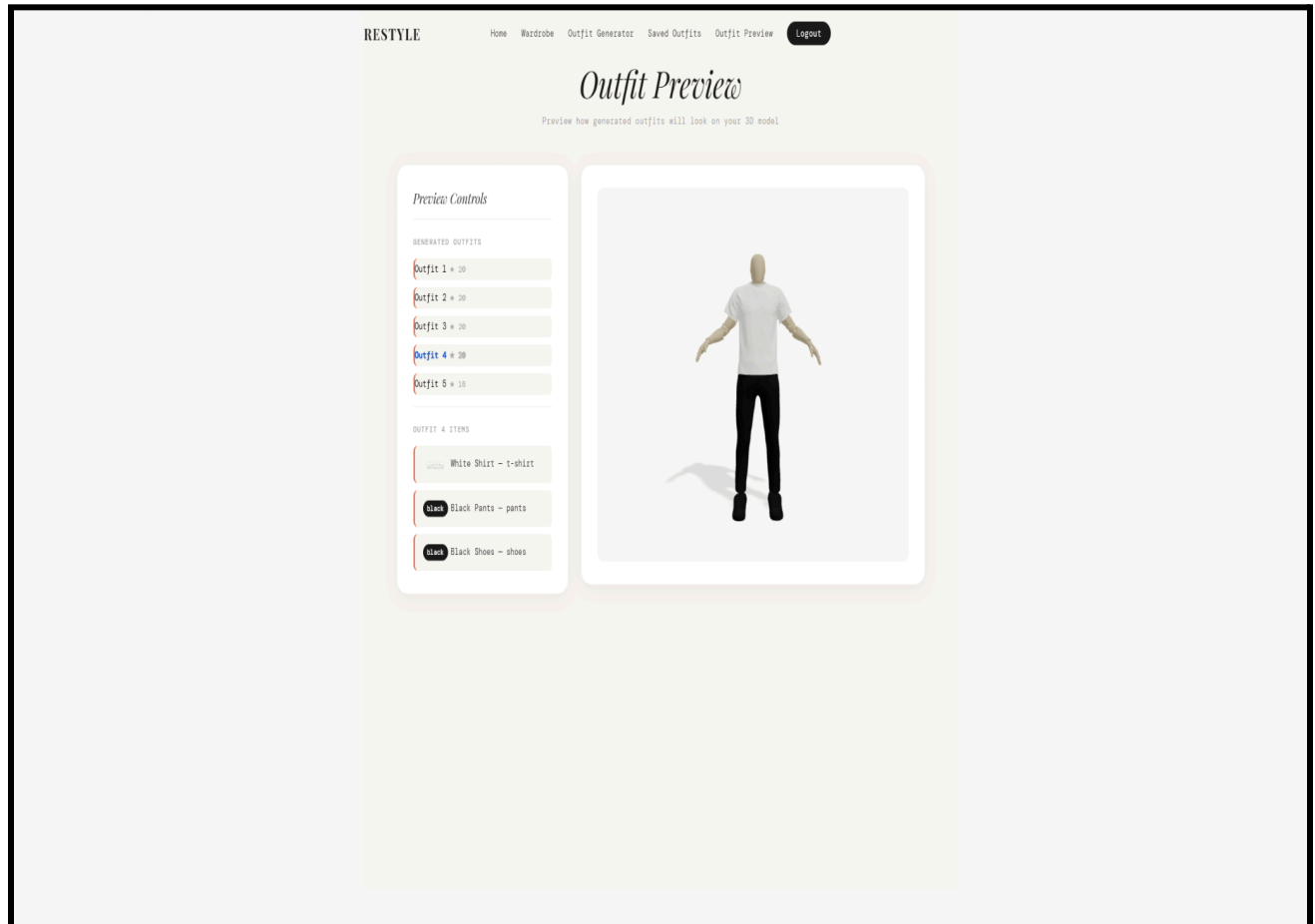


Figure 12 Saved Outfits (Option 1 Score 20)

10.7.1 Outfit Preview

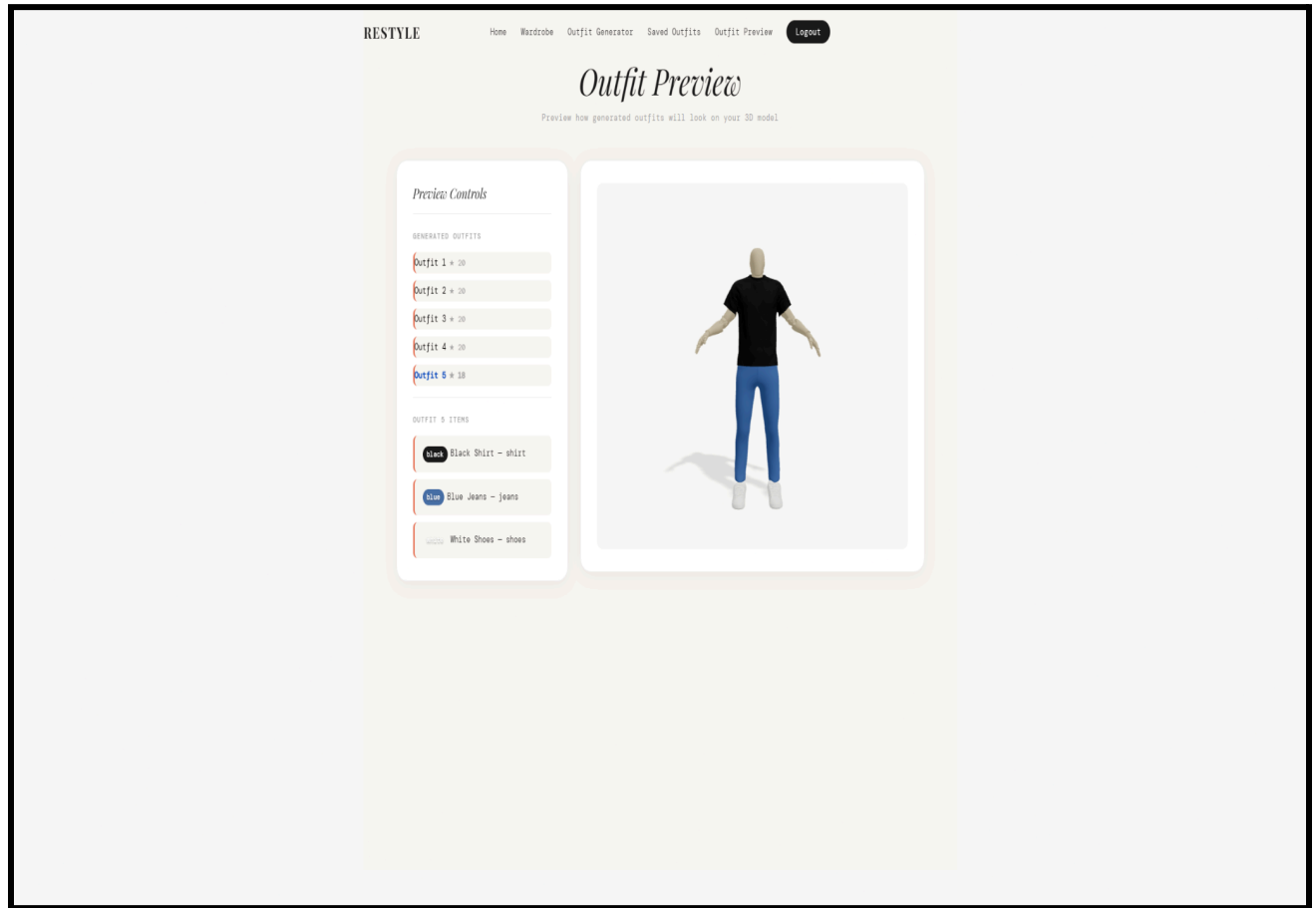


Figure 13 Saved Outfits (Option 2 Score 18)

11.0 Testing and Integration

Testing will be carried out to verify that all aspects of Restyle work together and individually. Unit tests will target the logic in the backend related to features such as authentication, wardrobe actions, and outfit generation algorithms. Integration tests will ensure that all components work well together by testing the interaction between frontend, backend, database, and scanning service with regard to features such as image uploads and scans. Specifically, it will be confirmed whether wardrobe items are persisted and fetched successfully, uploaded images are stored and rendered correctly, and scanning responses are formatted appropriately to display within the wardrobe form. Tests will also be made for the outfit generation feature to ensure that the correct data from stored wardrobe items, filter selections, and compatibility scores are being passed through the algorithm. Testing will be carried out for the complete user workflow from login to previewing outfits. Edge case testing will be done to ensure that scenarios such as wrong logins, lack of necessary form input fields, unsupported file types, scanning responses unavailability, empty wardrobes, and small sets of wardrobe items for outfit generation are considered.

12.0 Installation Instructions

To run the project locally, the user will install Node.js for the Express backend and React frontend, PostgreSQL for the database, and Python for the FastAPI scanning service. After cloning the repository, the user will configure environment variables (database connection string, JWT secret, and service URLs), install dependencies for both frontend and backend, and initialize the database schema. The backend server will be started first, followed by the frontend development server. If the ML scanning service is enabled, the user will also install Python dependencies and start the FastAPI server; otherwise, the application will still support manual clothing entry.

13.0 Recommendations for Enhancement

After the MVP is complete, Restyle can be improved in several ways. The scanning feature could be expanded to predict more attributes such as patterns, materials, and style tags, and accuracy could be improved through better training data or fine-tuning. Outfit generation could become more personalized by learning from saved outfits or user ratings and could factor in the user's weather so that outfit suggestions are appropriate for the current conditions. It can also be improved by making accessories such as watches, hats, bags, and jewelry as part of the outfit combinations. The preview feature could be enhanced with better mannequin customization, allowing users to adjust the mannequin's body type and height to give a more realistic clothing placement. A virtual try-on feature can be added that will make an image of the user wearing the clothing items instead of the mannequin, giving a more accurate preview of how an outfit would actually look on them. The website can also integrate social features that lets the users have the ability to share their outfits with friends or post in public where other users can see. These enhancements would build on the current foundation and make the application more advanced and useful over time.

14.0 References

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